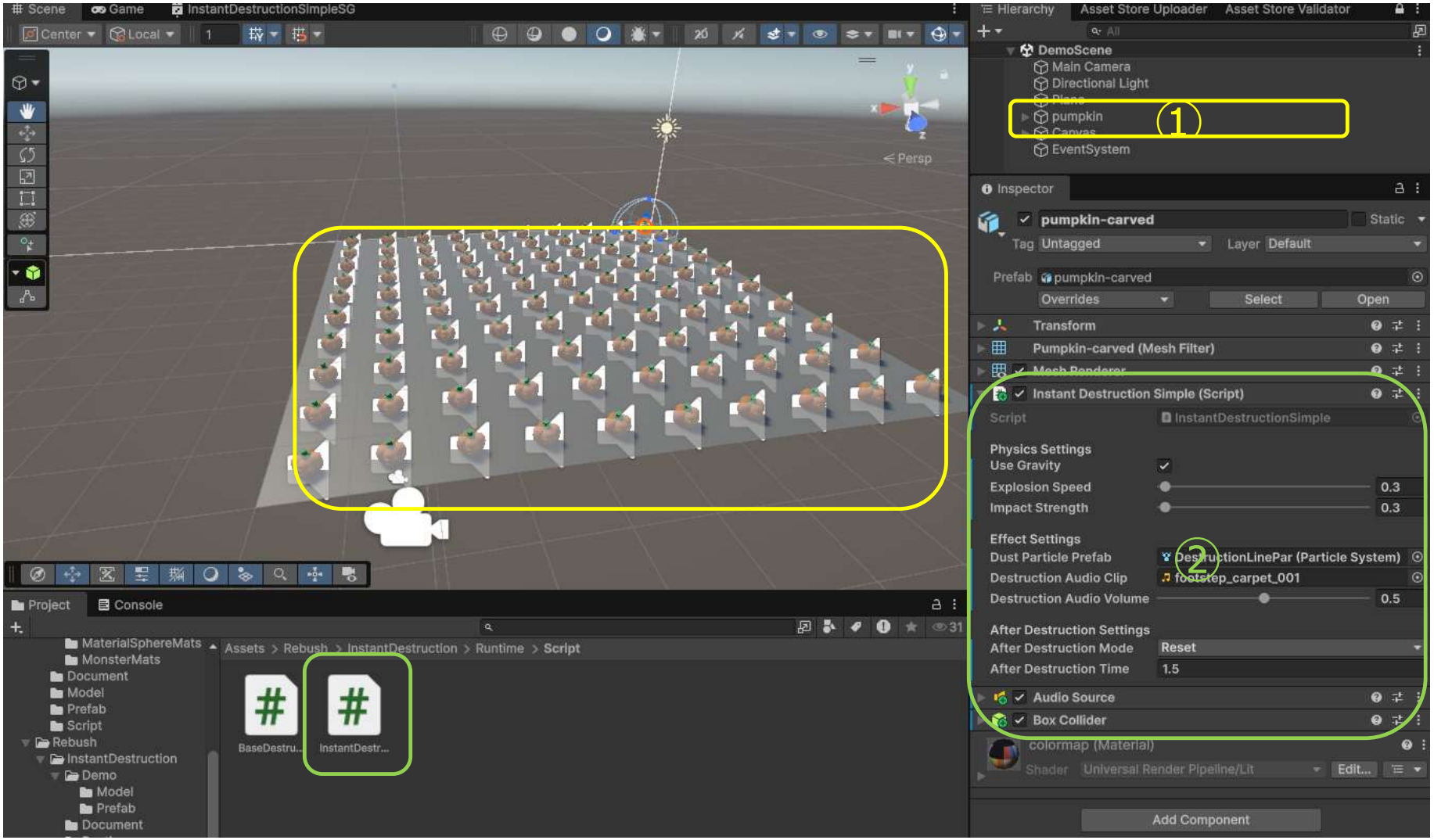
How To Use Instant Destruction

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1. DemoScene



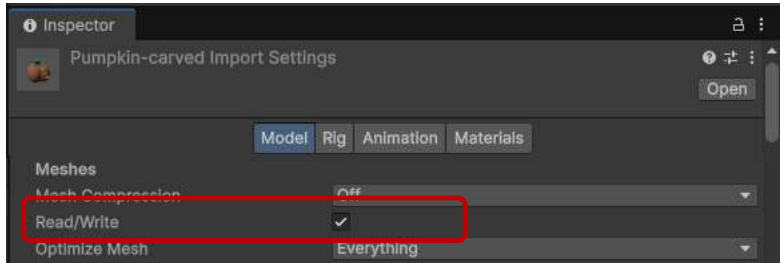
①Demo Gameobject: pumpkin
②Instant Destruction Component

The placement may change due to updates or changes.

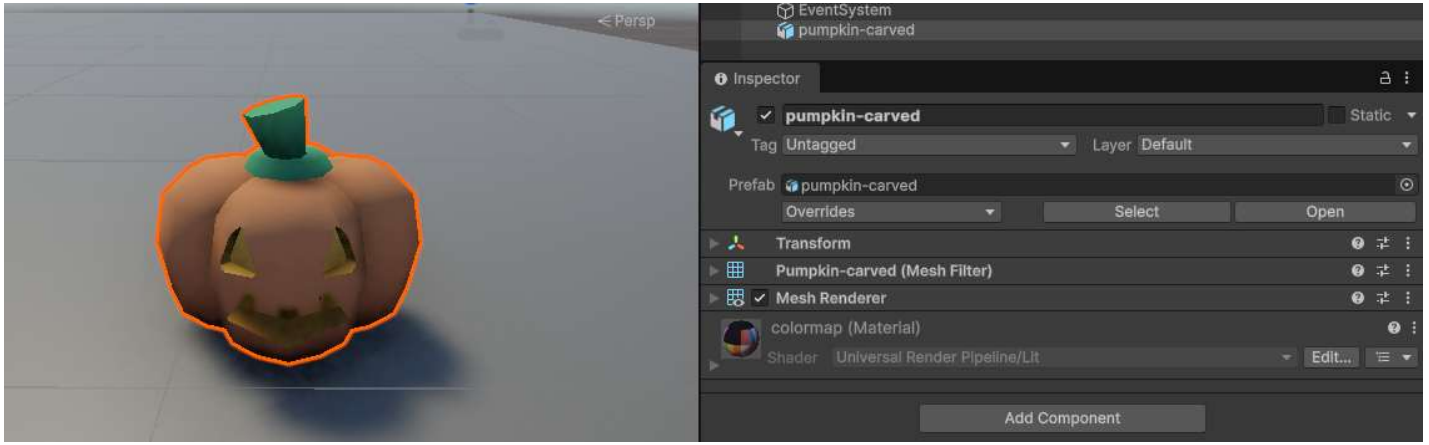
2. Quick Start (3 Steps)

Step 1: Model Setup

Prepare the model and enable the "Read/Write" property.

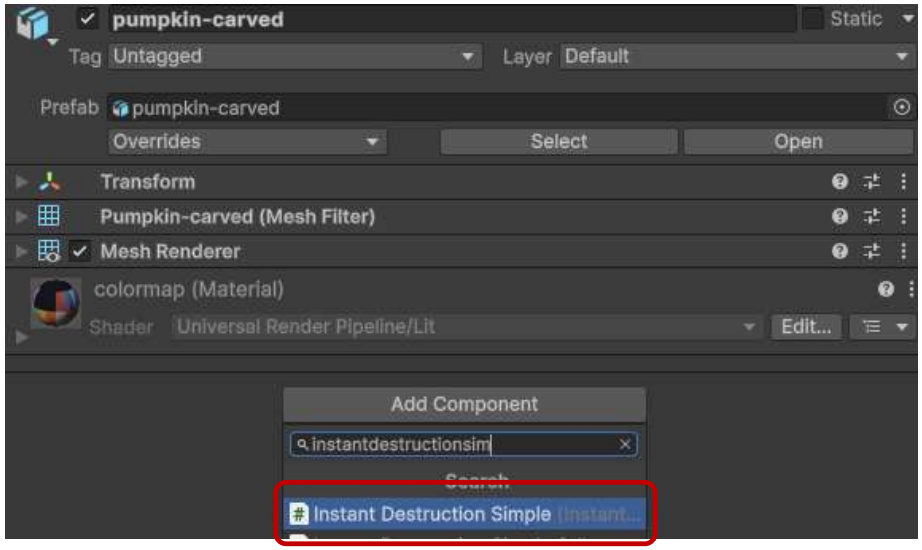


and Instantiate the model.



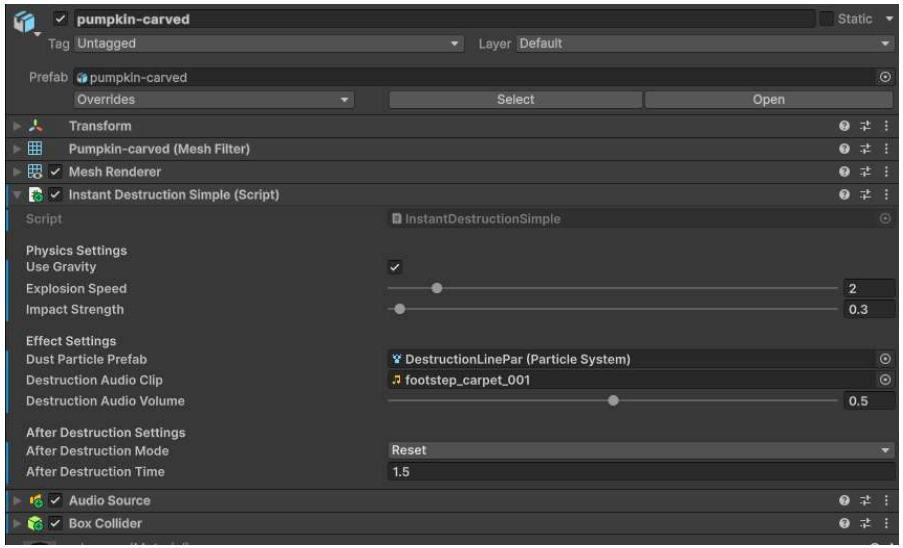
Step 2: Component Setup

Attach the "Instant Destruction Simple" component.



※The baseline component.
Swap or add components to switch between different destruction styles.

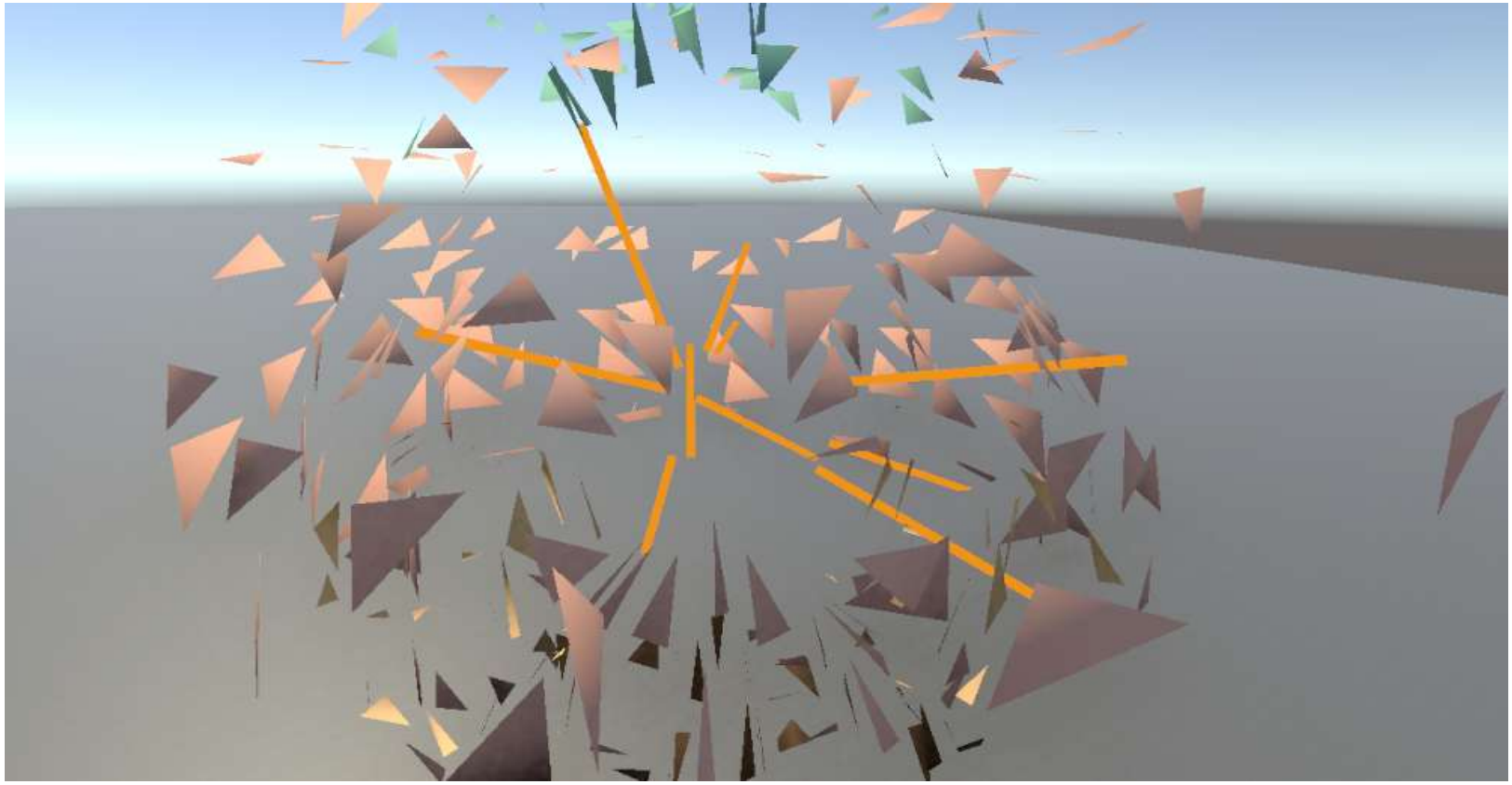
Ready to use with default settings. Customize as needed.



※AudioSource and BoxCollider are added automatically.

Step 3: Let's test it out!

Trigger the destruction via collider collision, or call the method from your script.



3. Notes / Requirements

※ Specifications, class structures, and limitations are subject to change in future updates.

1. Mesh Setup (Required)

Read/Write Enabled:

The mesh must have Read/Write enabled (isReadable = true)
in its import settings. If disabled, a warning will appear and destruction will fail.

Required Components:

The GameObject must have a MeshFilter and a Renderer (e.g., MeshRenderer) attached.

2. Requirements & Limitations

GPU Calculation:

Uses Compute Shaders and Graphics.RenderMeshIndirect.
The target platform and Graphics API must support these features.

No Fragment Shadows:

To optimize performance,
shadows for shattered fragments are disabled (ShadowCastingMode.Off).

3. Collision & Collider Behavior

Collision Trigger:

Destruction triggers automatically via OnCollisionEnter.
The center of the explosion is calculated from the average of all contact points.

Auto-Collider:
If no collider is present,
a BoxCollider matching the mesh size is automatically added at startup.

State Control: * On Destruction:
The collider is disabled (enabled=false) so characters/projectiles can pass through fragments.

On Reset:
The collider is automatically re-enabled (enabled = true).

4. After Destruction Mode
Configure the object's behavior after destruction via afterDestructionMode in the Inspector:

Nothing: Keeps the fragments scattered on the ground.

Destroy: Completely destroys the GameObject after afterDestructionTime (seconds).

Reset: Resets the object to its original state after the timer, making it ready to be broken again.

5. Performance & System Details
Shader Warmup:
To prevent frame drops from async shader compilation,
a dummy render (scale 0) runs off-screen for the first 120 frames at startup.

Auto-AudioSource:

An AudioSource component is automatically added if not found, used for playing the destruction sound.

Auto-Scaling FX:

Dust particles (dustParticlePrefab) automatically scale between 1.0 and 4.0 based on the object's original size.